
3AL Test 2 2022 Solutions

Time: 90 minutes

Available marks: 80

Maximum marks: 75

Instructions

1. Answer all questions in the lined answer booklets.
 2. Start each question on a new page.
 3. Make sure each answer is clearly labelled.
 4. Show all work and reasoning.
 5. This test consists of **4** questions. Make sure your question paper is complete.
 6. If you've read the instructions write your favourite Greek letter on the front of your booklet :) Good luck! Extra mark if you do
-

1. (a) State the definition of a group action.

Definition (Group action). Let G be a group and X a set. A **group action** by G on X is a map $*$: $G \times X \rightarrow X$ satisfying the following properties:

- (i) Compatibility: $g_1 * (g_2 * x) = (g_1 g_2) * x$ for all $g_1, g_2 \in G$ and $x \in X$,
- (ii) Identity: $1 * x = x$ for all $x \in X$.

Notes

Map [✓].

Compatibility [✓²].

Identity [✓²].

- (b) State the definition of a stabiliser.

Definition (Stabiliser). Let G be a group acting on a set X . Then the **stabiliser** of an element $x \in X$ is

$$S(x) = \{g \in G : g * x = x\}.$$

Notes

- Suffices to write “ $S(x) = \dots$ ”.

(c) Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism and let $*_2 : G_2 \times X \rightarrow X$ be a group action by G_2 on X .

(i) Prove that the map $*_1 : G_1 \times X \rightarrow X$ defined by $g *_1 x = \varphi(g) *_2 x$ is a group action.

(ii) Let $S_1(x)$ denote the stabiliser of $x \in X$ under the group action $*_1$. Show that $\ker \varphi \subseteq S_1(x)$ for all $x \in X$.

(i) Take $g, g' \in G_1$ and $x \in X$. Then

$$\begin{aligned}(gg') *_1 x &= \varphi(gg') *_2 x = \varphi(g)\varphi(g') *_2 x \\ &= \varphi(g) *_2 (\varphi(g') *_2 x) = \varphi(g) *_2 (g' *_1 x) \\ &= g *_1 (g' *_1 x).\end{aligned}$$

Moreover,

$$1_{G_1} *_1 x = \varphi(1_{G_1}) *_2 x = 1_{G_2} *_2 x = x.$$

Notes

Compatibility [\[✓³\]](#).

Identity [\[✓²\]](#).

(ii) Let $g \in \ker \varphi$. Then $\varphi(g) = 1_{G_2}$. This means that

$$g *_1 x = \varphi(g) *_2 x = 1_{G_2} *_2 x = x.$$

That is, $g \in S_1(x)$ and so $\ker \varphi \subseteq S_1(x)$.

Notes

- 2 marks for saying $\varphi(g) = 1_{G_2}$ (can earn the marks if identity is written as 1 instead of 1_{G_2}).

[5 + 5 + 5 + 5 = 20 marks]

NOTE: Be careful with notation in (c). Do not confuse $*_1$ with $*_2$, etc.

2. (a) State the Class Equation.

Class Equation. Let G be a finite group and class a_1 , class a_2 , ..., class a_n be the distinct nonsingleton conjugacy classes. Then

$$|G| = |Z(G)| + \sum_{i=1}^n |G : C_G(a_i)|.$$

Notes

- Lose 2 marks for not defining a_i s.

(b) State the definition of a p -group and give an example.

Definition (p -Group). Let p be a prime. A group G is called a p -group if $|G| = p^n$ for some $n \geq 1$.

Any valid example, e.g. $\mathbb{Z}_2$, $\mathbb{Z}_3$, $\mathbb{Z}_9$, K_4 , D_4 .

Notes

Definition [✓³].

- Alright to use definition in terms of orders of elements.

Example [✓²].

(c) Let G be a p -group. Prove that $Z(G)$ is nontrivial.

In the Class Equation, each index divides $|G|$, which is p^n , for some $n \geq 1$. Thus, each index must equal p^k for some k satisfying $1 \leq k \leq n$. Note that we cannot have $k = 0$ since that would imply $|G : C_G(a_i)| = 1$ and so $C_G(a_i) = G$, that is, $a_i \in Z(G)$, which contradicts a_i being a nonsingleton conjugacy class.

So, p divides $|G|$ and each index $|G : C_G(a_i)| = |G|/|C_G(a_i)|$. Thus, p must also divide $|Z(G)|$. Therefore, $|Z(G)| > 1$ (it is impossible for $|Z(G)| = 0$) and so $Z(G)$ is nontrivial.

Notes

p divides each index and $|G|$ [✓⁵].

- Lose 2 marks for not explaining why index cannot be 1.

p divides $|Z(G)|$ and so $Z(G) \neq \{1\}$ [✓⁵].

- Not necessary to explain why $|Z(G)| \neq 0$.
- Suffices to state that $|Z(G)| > 1$ instead of $Z(G) \neq \{1\}$.

[5 + 5 + 10 = 20 marks]

3. (a) State the Second Sylow Theorem.

Second Sylow Theorem. *Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Any two Sylow p -subgroups are conjugate.*

Notes

- Lose 1 mark for not indicating p divides $|G|$.

(b) Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Let H denote the intersection of all Sylow p -subgroups of G .

- Show that if P is a Sylow p -subgroup of G and $g \in G$, then gPg^{-1} is a Sylow p -subgroup of G .
- Prove that $|H|$ is a power of p .
- Prove that H is normal in G .

(i) Note that $|gPg^{-1}| = |P| = p^n$. Moreover, gPg^{-1} is a subgroup of G . Thus it is a Sylow p -subgroup.

Notes

- Not necessary to prove gPg^{-1} is a subgroup of G .
- Not necessary to prove $|gPg^{-1}| = |P|$.

(ii) By Lagrange's Theorem, the order of H must divide the order of each Sylow p -subgroup. That is, $|H| \mid p^n$. However, the only divisors of p^n are powers of p .

(iii) Take $g \in G$, we will show that $gHg^{-1} \subseteq H$. For any Sylow p -subgroup P of G , we have $H \subseteq P$. Then, for all $a \in G$, the conjugate aPa^{-1} contains H , as each conjugate is another Sylow p -subgroup of G by (i). Taking $a = g^{-1}$, we have $H \subseteq g^{-1}Pg$. Thus $gHg^{-1} \subseteq g(g^{-1}Pg)g^{-1} = P$, for all Sylow p -subgroups P . Thus gHg^{-1} is contained in every Sylow p -subgroup and so it is contained in the intersection, H . Then by the Normality Test, $H \triangleleft G$.

Notes

(Still thinking about mark allocation)

[5 + 5 + 5 + 10 = 25 marks]

4. (a) State the Third Sylow Theorem.

Third Sylow Theorem. Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Let n_p denote the number of Sylow p -subgroups of G . Then $n_p \equiv 1 \pmod{p}$ and n_p divides m .

Notes

- Lose 2 marks for not indicating $|G| = p^n m$ and $p \nmid m$ (okay to omit $n \geq 1$).
- Not necessary to define n_p .

(b) Prove that a group of order 20 has a normal subgroup.

Let G be a group of order $20 = 2^2 \times 5$. By the Third Sylow Theorem, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 4$. Thus $n_5 = 1$ and so there is a unique Sylow 5-subgroup of G . Moreover, since it is unique, the Sylow 5-subgroup is normal.

Notes

Show that $n_5 = 1$ [✓⁵].

- Lose 3 marks for not saying $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 4$.

Unique implies normal [✓⁵].

[5 + 10 = 15 marks]